Unit 7.01: Triangle Centers, Ceva, Menelaus & Stewart Theorem

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 18x + 98 in the domain of Triangle Centers, Ceva, Menelaus & Stewart. By completing the square f_1x = x - h^2 + k , find the sum k + f_111 .

[ ] A: 47
[ ] B: 21
[x] C: 38
[ ] D: 36

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Complete the square: x^2 - 18x + 98 = x - 9^2 - 81 + 98 = x - 9^2 + 17 , giving minimum value k = 17 . **[M1]** 2. Evaluate at x = 11 : f_111 = 2^2 + 17 = 21 . **[M1]** 3. Sum the two quantities: k + f_111 = 17 + 21 = 38 . **[A1]** - **Final Verified Answer:** 38 .

Q2. A mathematical model in Triangle Centers, Ceva, Menelaus & Stewart is defined by the cubic function y_2x = 2x^3 - 9x^2 + 15x - 14 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 69
[ ] B: 0
[ ] C: 15 + 1
[x] D: 15

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 6x^2 - 18x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 69 - 183 + 15 = 15 . **[M1][A1]** - **Final Verified Answer:** 15 .

Q3. In Triangle Centers, Ceva, Menelaus & Stewart, two displacement vectors are given by mathbf{u}_{3} = langle 9, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[x] A: 170
[ ] B: 25
[ ] C: 145
[ ] D: 282

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 99 + -87 = 81 + -56 = 25 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 9^2 + -8^2 = 145 . **[M1]** 3. Combine: 25 + 145 = 170 . **[A1]** - **Final Verified Answer:** 170 .

Q4. The linear growth profile shown in the diagram for Triangle Centers, Ceva, Menelaus & Stewart follows an arithmetic progression with first term a_1 = 10 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 94
[x] B: 780
[ ] C: 870
[ ] D: 1410

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 10, d = 6 : S_{15} = frac{15}{2}[20 + 14 times 6] = frac{15}{2} times 104 = 780 . **[M1][A1]** - **Final Verified Answer:** 780 .

Unit 7.01: Triangle Centers, Ceva, Menelaus & Stewart Theorem

Q5. In Triangle Centers, Ceva, Menelaus & Stewart, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 36x,mathrm{d}x .

[ ] A: 144
[ ] B: 156
[x] C: 120
[ ] D: 61

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Integrate term by term: int 18x^2 + 36x,mathrm{d}x = left[6x^3 + 18x^2right]_0^{2} . **[M1]** 2. Substitute the upper and lower limits: 62^3 + 182^2 - 0 = 48 + 72 = 120 . **[M1][A1]** - **Final Verified Answer:** 120 .

Q6. In Triangle Centers, Ceva, Menelaus & Stewart, consider the quadratic function f_6x = x^2 - 20x + 189 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_617 .

[ ] A: 237
[ ] B: 138
[ ] C: 220
[x] D: 227

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Complete the square: x^2 - 20x + 189 = x - 10^2 - 100 + 189 = x - 10^2 + 89 , giving minimum value k = 89 . **[M1]** 2. Evaluate at x = 17 : f_617 = 7^2 + 89 = 138 . **[M1]** 3. Sum the two quantities: k + f_617 = 89 + 138 = 227 . **[A1]** - **Final Verified Answer:** 227 .

Q7. The graph illustrates the cubic model y_7x = 2x^3 - 10x^2 + 33x - 21 studied in Triangle Centers, Ceva, Menelaus & Stewart. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[x] A: 257
[ ] B: 417
[ ] C: 224
[ ] D: 61

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 6x^2 - 20x + 33 . **[M1]** 2. Substitute x = 8 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=8} = 664 - 208 + 33 = 257 . **[M1][A1]** - **Final Verified Answer:** 257 .

Q8. In Triangle Centers, Ceva, Menelaus & Stewart, two displacement vectors are given by mathbf{u}_{8} = langle 10, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 45
[x] B: 194
[ ] C: 149
[ ] D: 264

Solution: **Mark Scheme & Examiner Solution Triangle Centers, Ceva, Menelaus & Stewart:** 1. Compute the inner product: mathbf{u}_{8} cdot mathbf{v}_{8} = 108 + -75 = 80 + -35 = 45 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{8}|^2 = 10^2 + -7^2 = 149 . **[M1]** 3. Combine: 45 + 149 = 194 . **[A1]** - **Final Verified Answer:** 194 .